Introduction to Deep Learning

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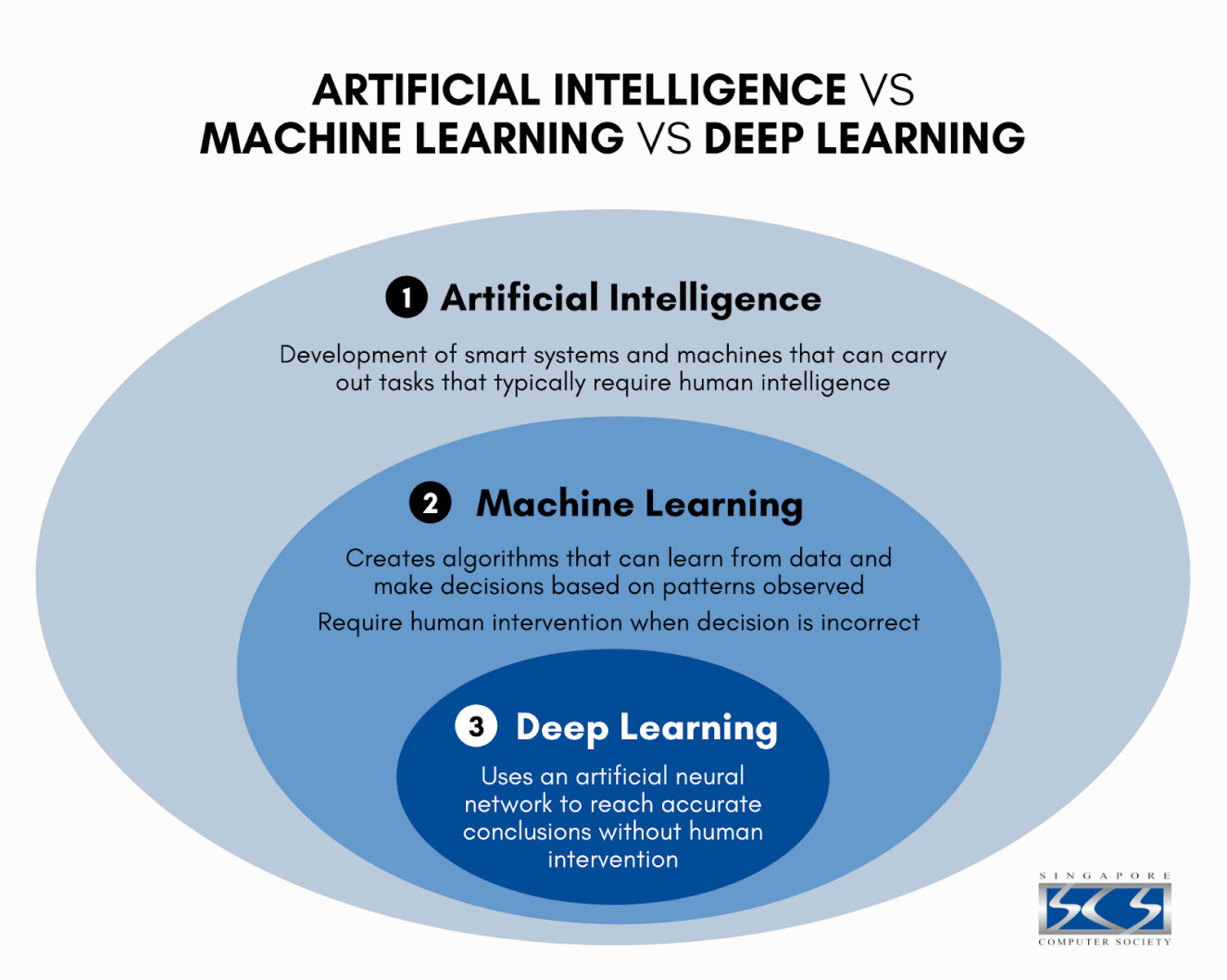
AI, ML, and DL Hierarchy



AI (broad goal) $\supset$ **Machine Learning** (data-driven) $\supset$ **Deep Learning** (neural networks)

What is Deep Learning?

- A subfield of Machine Learning focused on learning complex patterns using **multi-layer neural networks**.
- Inspired by the structure and function of the human brain.
- Learns hierarchical representations from raw data (images, text, audio).
- Powers modern AI: GPT, BERT, self-driving cars, medical AI, robotics.

Why Deep Learning Matters Today

- Dominates state-of-the-art in NLP, speech, and reinforcement learning.
- Enables automatic feature extraction.
- Scales with data and computation.
- Forms the foundation of large foundation models (LLMs, Vision Transformers).
- DL models have achieved human-level or superhuman performance in several domains.

Applications of Deep Learning

- Medical imaging and diagnostics
- Fraud detection and finance
- Climate modeling and remote sensing
- Natural language processing (GPT, BERT)
- Civic applications like governance dashboards, citizen sentiment analysis

History and Milestones

- 1943 – McCulloch & Pitts neuron
- 1958 – Rosenblatt's Perceptron
- 1986 – Backpropagation (Rumelhart et al.)
- 2012 – AlexNet revolution in vision
- 2017 – Transformers for sequence modeling
- 2020s – Foundation models and LLMs

Deep Learning vs. Traditional Machine Learning

Aspect	**Traditional Machine Learning**	**Deep Learning**
Feature Engineering	Requires hand-crafted features designed by experts	Learns hierarchical features automatically from raw data
Data Requirements	Works well with small–medium sized structured datasets	Typically needs very large datasets, especially for vision/NLP
Model Complexity	Simpler models (e.g., linear/logistic regression, SVM, trees)	Very high-capacity models (deep neural networks with many layers)
Computational Cost	Can run on standard CPUs; training usually fast	Often requires GPUs/TPUs; training can be computationally expensive
Interpretability	Often more interpretable (e.g., linear models, shallow trees)	Frequently "black-box"; requires XAI methods (SHAP, LIME, etc.)
Typical Data Types	Mainly structured/tabular data, simple time series	Images, text, audio, video, complex high-dimensional

Mathematical Perspective

- Goal: approximate a complex function $f^*(x)$.
- Neural networks define $f(x;\theta)$ and learn parameters θ minimizing loss $L(y,\hat{y})$.
- Universal approximation theorem:

$$\forall f^*, \exists\ f(x;\theta) \text{ s.t. } |f(x;\theta) - f^*(x)| < \epsilon$$

The Artificial Neuron (Perceptron)

A perceptron is a *simple neuron*. A multilayer neural network is a *more powerful, differentiable perceptron* used for deep learning.
The perceptron models a single decision unit:

$$y = f(\mathbf{w}^T x + b)$$

- x: input vector
- w: weights
- b: bias
- f: activation function

What is the Bias in a Neuron?

A typical neuron computes:

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

- w_i are the **weights**.
- b is the **bias term**.
- The activation is then: $a = \sigma(z)$.

Intuition: The bias shifts the input to the activation function, giving the neuron more flexibility.

Purpose of Bias in Neural Networks

- **Shifts the activation:**
  Bias allows the neuron to move the activation function left or right.

$$z = w^\top x + b$$

  Without b, the neuron output is constrained to pass through the origin.

- **Learns the intercept:**
  Analogous to the intercept term in linear regression:

$$y = wx + b$$

- **Increases model flexibility:**
  Enables the network to represent functions that are not forced to be zero at $x = 0$.

Bias = small parameter but has huge impact on model capacity.

From Single Neuron to Neural Networks

- A neural network stacks many neurons in layers.
- Information flows from input $\rightarrow$ hidden $\rightarrow$ output.
- Deeper layers learn more abstract features.

$$y = f_L(W_L f_{L-1}(\cdots f_1(W_1 x)))$$

Hierarchical Feature Learning

Example: Image Processing

- Layer 1: Edges
- Layer 2: Shapes
- Layer 3: Object parts
- Layer 4: Objects (faces, cars)

Deep learning learns these automatically.

Mathematical Deep Dive: Perceptron vs. Modern Neuron

1. Perceptron Model (Rosenblatt, 1957)

$$z = \sum_{i=1}^{n} w_i x_i + b$$

$$\hat{y} = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$$

Characteristics

- Uses a **Heaviside step function** $H(z)$.
- Output is strictly binary.
- **Not differentiable** $\rightarrow$ cannot use gradient descent.
- Can only learn **linearly separable** decision boundaries.

2. Modern Artificial Neuron (Deep Learning)

$$z = \sum_{i=1}^{n} w_i x_i + b$$

$$\hat{y} = \sigma(z)$$

Where $\sigma(z)$ is a differentiable activation function:

Weights, Loss, and Learning

- **Weights**: parameters the network learns.
- **Loss function**: measures error.

$$L = (y - \hat{y})^2 \quad \text{(example)}$$

- **Gradient descent**: updates weights to reduce loss.
- **Backpropagation**: computes gradients using chain rule.

Gradient Descent Intuition

Move weights in the direction that decreases error:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

- η: learning rate
- The network learns by adjusting weights over many iterations.

Learning Rate in Deep Learning

Definition

- The **learning rate** (denoted η) is a hyperparameter that controls how large the weight updates are during gradient descent.

Mathematical Role

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial \mathcal{L}}{\partial w}$$

Interpretation

- η determines how **fast or slow** a model learns.
- Too large η: updates overshoot $\rightarrow$ training may **diverge**.
- Too small η: updates are tiny $\rightarrow$ training becomes **very slow**.

Why It Matters

- A good learning rate accelerates convergence.
- It stabilizes training and prevents oscillations.

Key Insight: The learning rate is NOT an activation function. It is a **training hyperparameter** that controls gradient descent behavior.

Structure of FFNN

- Input layer: raw features
- Hidden layers: nonlinear transformations
- Output layer: predictions (class scores, regression output)

$$\hat{y} = f(x) = W_3\sigma(W_2\sigma(W_1x + b_1) + b_2) + b_3$$

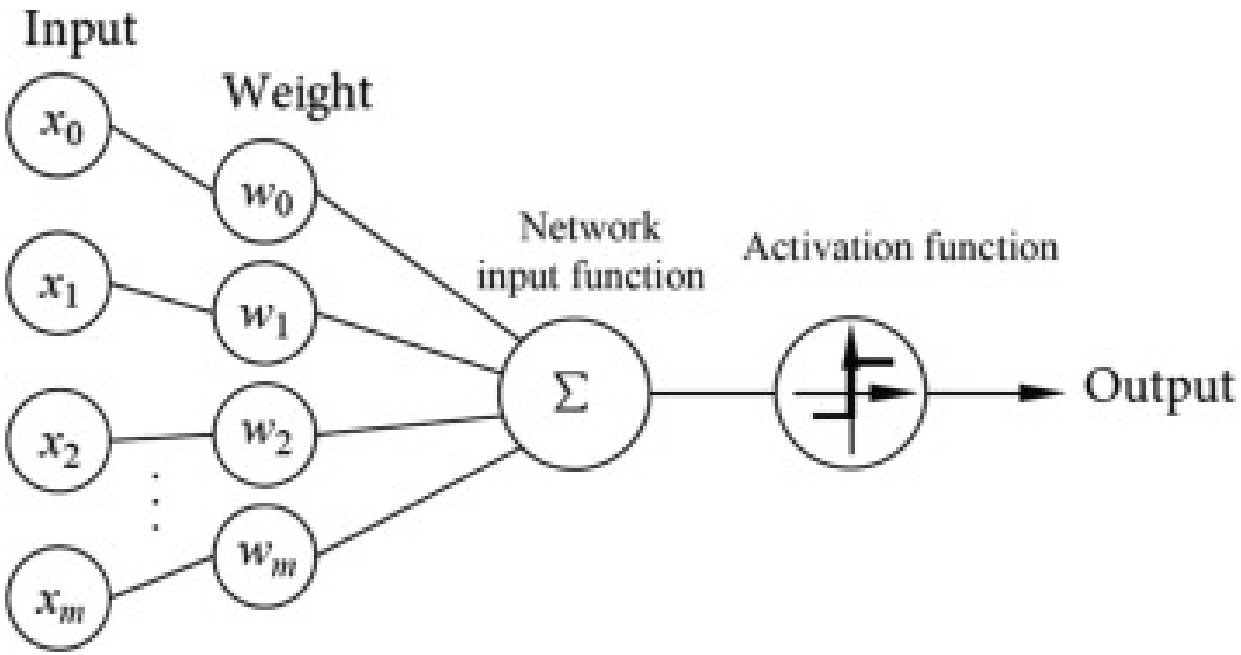
Representation Learning

Each hidden layer learns progressively abstract representations.

Activation Functions

Function	Formula
Sigmoid	$\frac{1}{1+e^{-x}}$
Tanh	$\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$
ReLU	$\max(0, x)$
Leaky ReLU	$\max(0.01x, x)$
GELU	$x * \frac{1}{2}[1+\mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)]$ where $\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt$

Structure of a Perceptron



- Inputs: $x_1, x_2, ..., x_n$
- Weights: $w_1, w_2, ..., w_n$
- Bias: b

Training the Perceptron

Weights are updated using the **Perceptron Learning Rule**:

$$w_i \leftarrow w_i + \eta(y - \hat{y})x_i$$

$$b \leftarrow b + \eta(y - \hat{y})$$

Where:

- η = learning rate
- y = true label
- $\hat{y}$ = predicted label

What Perceptrons Can Do

Strengths

- Solve linearly separable problems
- Perform logical operations (AND, OR)
- Form the base of Multi-Layer Perceptrons (MLPs)

What Perceptrons Cannot Do

Limitations

- Cannot solve non-linear problems like XOR
- Only outputs 0 or 1 (no probabilities)
- Limited expressive power

This leads to Multi-Layer Perceptrons and Deep Learning

Exercise : Network Specification

Consider a neural network with one hidden neuron.

- Input:

$$x = [2.0,\ -1.0]$$

- Weights and biases:

$$W_1 = [0.5 \quad -0.3], \qquad b_1 = 0.1$$

$$W_2 = 0.8, \qquad b_2 = -0.2$$

- Activation: Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Target:

$$y = 1$$

- Loss:

$$L = \frac{1}{2}(\hat{y} - y)^2$$

- Learning rate:

$$\alpha = 0.1$$

Tasks (Forward Pass)

Compute:

1. Hidden pre-activation:
$$z_1 = W_1 x + b_1$$
2. Hidden activation:
$$a_1 = \sigma(z_1)$$
3. Output pre-activation:
$$z_2 = W_2 a_1 + b_2$$
4. Network output:
$$\hat{y} = \sigma(z_2)$$
5. Loss:
$$L = \frac{1}{2}(\hat{y} - y)^2$$

Backpropagation Tasks

1. Compute:
$$\frac{\partial L}{\partial \hat{y}}, \qquad \frac{\partial L}{\partial z_2}$$
2. Gradients:
$$\frac{\partial L}{\partial W_2}, \qquad \frac{\partial L}{\partial b_2}$$
3. Hidden layer:
$$\frac{\partial L}{\partial a_1}, \qquad \frac{\partial L}{\partial z_1}$$
4. Hidden parameters:
$$\frac{\partial L}{\partial W_1}, \qquad \frac{\partial L}{\partial b_1}$$

Gradient Descent Updates

Update the parameters:

$$W_i^{new} = W_i - \alpha \frac{\partial L}{\partial W_i}, \qquad b_i^{new} = b_i - \alpha \frac{\partial L}{\partial b_i}$$

Interpretation questions:

- What happens if W_2 is very large?
- What is the effect of very small or large learning rate?

Deep Learning Workflow

1. Data collection & preprocessing
2. Model design & hyperparameters
3. Training and validation
4. Evaluation and interpretability
5. Deployment and monitoring

Hyperparameter Tuning

- Learning rate, batch size, optimizer
- Number of layers, units per layer
- Activation and dropout rate
- Automated tuning tools: Grid search, Random search, Bayesian optimization

Gradient Descent: Objective & Update Rule

Learning Goal:

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{m} \sum_{i=1}^{m} \ell(f_\theta(x^{(i)}), y^{(i)})$$

Gradient Descent Update:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta \mathcal{L}(\theta^{(t)})$$

For a single weight w_j:

$$w_j^{(t+1)} = w_j^{(t)} - \eta \frac{\partial \mathcal{L}}{\partial w_j}$$

Key Challenge: Neural networks are nested functions $\Rightarrow$ we must apply the **multivariable chain rule**.

Chain Rule in Deep Networks

Neural network computations:

$$x \to z^{(1)} \to a^{(1)} \to z^{(2)} \to a^{(2)} \to \hat{y}$$

Chain Rule (general form):

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

For neural networks:

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}$$

This principle extends to all layers $\Rightarrow$ **backpropagation**.

MLP Forward Pass (Two-Layer Network)

Layer 1 (Hidden Layer)

$$z^{(1)} = W^{(1)}x + b^{(1)}$$
$$a^{(1)} = \sigma(z^{(1)})$$

Layer 2 (Output Layer)

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$$
$$\hat{y} = \sigma(z^{(2)})$$

Loss (MSE):

$$\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2$$

Backpropagation: Output Layer

We compute

$$\frac{\partial \mathcal{L}}{\partial W^{(2)}} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial b^{(2)}}$$

Step 1: Output layer error term

$$\delta^{(2)} = \frac{\partial \mathcal{L}}{\partial z^{(2)}} = (\hat{y} - y) \cdot \sigma'(z^{(2)})$$

Step 2: Gradients

Weights:

$$\frac{\partial \mathcal{L}}{\partial W^{(2)}} = \delta^{(2)} (a^{(1)})^T$$

Bias:

$$\frac{\partial \mathcal{L}}{\partial b^{(2)}} = \delta^{(2)}$$

This completes the gradient for the output layer.

Backpropagation: Hidden Layer

We compute

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial b^{(1)}}$$

Step 1: Hidden layer error term

$$\delta^{(1)} = \left((W^{(2)})^T \delta^{(2)} \right) \odot \sigma'(z^{(1)})$$

Step 2: Gradients

Weights:

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} = \delta^{(1)} x^T$$

Bias:

$$\frac{\partial \mathcal{L}}{\partial b^{(1)}} = \delta^{(1)}$$

Thus each layer receives error from above and multiplies by the activation derivative.

Final Gradient Descent Updates for MLP

Output Layer Updates

$$W^{(2)} \leftarrow W^{(2)} - \eta\, \delta^{(2)} (a^{(1)})^T$$
$$b^{(2)} \leftarrow b^{(2)} - \eta\, \delta^{(2)}$$

Hidden Layer Updates

$$W^{(1)} \leftarrow W^{(1)} - \eta\, \delta^{(1)} x^T$$
$$b^{(1)} \leftarrow b^{(1)} - \eta\, \delta^{(1)}$$

Key Insight: Backpropagation = repeated application of the chain rule across layers.
Outcome: MLPs learn by propagating error backwards and adjusting weights to reduce the loss.

Multilayer Perceptron (MLP) for Classification

Key idea: MLP learns hierarchical representations and uses Softmax for multi-class outputs.

- Input: feature vector x
- Hidden layers: nonlinear transformations
- Output layer: class scores
- Softmax converts scores to probabilities
- Loss: Cross-entropy

$$\text{class}(x) = \arg\max_k \hat{y}_k$$

MLP Architecture Example

- Input dimension: 4 (Iris features)
- Hidden layer: 5 neurons
- Activation: ReLU
- Output layer: 3 neurons
- Activation: Softmax

$$x \rightarrow \text{Hidden Layer} \rightarrow \text{Output Layer} \rightarrow \hat{y}$$

Step 1: Hidden Layer Computation

$$z_1 = W_1 x + b_1$$

Activation (ReLU):

$$a_1 = \max(0, z_1)$$

ReLU introduces non-linearity.

Step 2: Output Layer Computation

$$z_2 = W_2 a_1 + b_2$$

Softmax converts scores to probabilities:

$$\hat{y}_k = \frac{e^{z_{2k}}}{\sum_j e^{z_{2j}}}$$

$$\hat{y} = [\hat{y}_1, \hat{y}_2, \hat{y}_3]$$

Step 3: Cross-Entropy Loss

True label (one-hot):

$$y = [y_1,\ y_2,\ y_3]$$

Loss:

$$L = -\sum_k y_k \log(\hat{y}_k)$$

For classification:

$$L = -\log(\hat{y}_{\text{correct class}})$$

Derivative at Output Layer

Softmax + Cross-Entropy simplifies:

$$\frac{\partial L}{\partial z_2} = \hat{y} - y$$

This gradient is used to update the output layer.

Gradients for Output Layer Parameters

$$\frac{\partial L}{\partial W_2} = (\hat{y} - y)\, a_1^T$$

$$\frac{\partial L}{\partial b_2} = \hat{y} - y$$

Gradient Flow to Hidden Layer

$$\frac{\partial L}{\partial a_1} = W_2^T(\hat{y} - y)$$

ReLU derivative:

$$\text{ReLU}'(z_1) = \begin{cases} 1 & z_1 > 0 \\ 0 & z_1 \leq 0 \end{cases}$$

Thus:

$$\frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial a_1} \odot \text{ReLU}'(z_1)$$

Gradients for Input Layer Parameters

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial z_1} x^T$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1}$$

Parameter Updates

Gradient descent:

$$W^{new} = W - \alpha \frac{\partial L}{\partial W}$$

$$b^{new} = b - \alpha \frac{\partial L}{\partial b}$$

Learning rate α controls the update step.

Iris Dataset (3-Class Classification)

Features:

- Sepal length
- Sepal width
- Petal length
- Petal width

Classes:

- Setosa
- Versicolor
- Virginica

MLP learns to separate these classes using learned features.

Classification Output Example

Example prediction:

$$\hat{y} = [0.95,\ 0.04,\ 0.01]$$

Meaning:

- 95% Setosa
- 4% Versicolor
- 1% Virginica

Predicted class = Setosa

Emerging Frontiers

- Self-supervised learning
- Multimodal AI (text+image+audio)
- Efficient architectures (MobileNet, TinyBERT)
- Quantum deep learning
- Civic and educational AI

Key Takeaways

- Deep learning automates hierarchical representation learning.
- Key architectures: FFNN, CNN, RNN/LSTM, Transformers, RL.
- The field blends theory, optimization, and ethics.
- Future: Explainable, energy-efficient, human-centered AI.

Mini Demo: XOR with PyTorch

```
import torch
import torch.nn as nn
import torch.optim as optim

X = torch.tensor([[0.,0.],[0.,1.],[1.,0.],[1.,1.]])
y = torch.tensor([[0.],[1.],[1.],[0.]])

model = nn.Sequential(
    nn.Linear(2, 2),
    nn.ReLU(),
    nn.Linear(2, 1),
    nn.Sigmoid()
)

criterion = nn.BCELoss()
optimizer = optim.Adam(model.parameters(), lr=0.1)

for epoch in range(2000):
    optimizer.zero_grad()
    y_pred = model(X)
    loss = criterion(y_pred, y)
    loss.backward()
    optimizer.step()

print(model(X))
```

Keras Example

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout

model = Sequential([
    Dense(128, activation='relu', input_shape=(X_train.shape[1],)),
    Dropout(0.3),
    Dense(64, activation='relu'),
    Dense(1, activation='sigmoid')
])
model.compile(optimizer='adam', loss='binary_crossentropy',
              metrics=['accuracy'])
model.fit(X_train, y_train, validation_data=(X_val, y_val),
          epochs=25, batch_size=32)
```

Exercise 1: Forward Pass Computation

Given: A 1-hidden-layer MLP

$$W_1 = \begin{bmatrix} 0.3 & -0.2 & 0.1 \\ 0.5 & 0.4 & -0.3 \end{bmatrix}, \quad b_1 = [0.1,\ -0.05]$$

$$W_2 = [0.4,\ -0.1], \quad b_2 = 0.2$$

Input:

$$x = [1.0,\ 0.5,\ -1.0]$$

Tasks:

1. Compute $z_1 = W_1 x + b_1$
2. Apply ReLU activation
3. Compute output $z_2 = W_2 a_1 + b_2$
4. Apply Sigmoid to get $\hat{y}$

Exercise 2: Backpropagation

Using the same MLP:

$$L = \frac{1}{2}(\hat{y} - y)^2, \qquad y = 1$$

Tasks:

1. Compute
$$\frac{\partial L}{\partial \hat{y}} \qquad \frac{\partial L}{\partial z_2}$$
2. Compute gradients:
$$\frac{\partial L}{\partial W_2}, \frac{\partial L}{\partial b_2}$$
3. Backpropagate to hidden layer:
$$\frac{\partial L}{\partial z_1}$$
4. Compute:
$$\frac{\partial L}{\partial W_1}, \frac{\partial L}{\partial b_1}$$

Exercise 3: Model Comparison

Compare **Logistic Regression**, **SVM**, and **MLP** for the Iris dataset.
Tasks:

1. Train all three models on the same train/test split.
2. Report:
   - Accuracy
   - Precision/Recall/F1
   - Confusion Matrix
3. Discuss:
   - Which model performed best?
   - How does dataset size affect each model?
   - What about interpretability vs predictive power?

Exercise 4: Practical Implementation

Implement in Python (PyTorch / Scikit-Learn):

1. MLP classifier for Iris (1 hidden layer)
2. Logistic regression classifier
3. SVM classifier (linear + RBF kernel)

Tasks:

- Compare training time
- Plot decision boundaries for 2 features
- Analyze misclassified samples

Challenge Exercise

Build a deeper MLP:

$$4 \rightarrow 10 \rightarrow 5 \rightarrow 3$$

Tasks:

1. Derive gradients for all layers
2. Train it on the Iris dataset
3. Compare performance to a shallow MLP

Goal: Understand the effect of depth on classification.

Thank You!

Questions and Group Discussion

Recommended Reading

- Goodfellow, Bengio, Courville (2016). *Deep Learning*.
- Chollet (2021). *Deep Learning with Python*.
- Bishop (2006). *Pattern Recognition and Machine Learning*.
- Online: TensorFlow, PyTorch.

References

Fleuret, F. (2024).
The Little Book of Deep Learning.
University of Geneva.

Goodfellow, I., Bengio, Y., Courville, A. (2016).
Deep Learning.
MIT Press, Cambridge.

Bishop, C. M., & Bishop, H. (2023).
Deep Learning: Foundations and Concepts.
Springer Nature.

Prince, S. J. (2023).
Understanding Deep Learning.
MIT Press.

Kelleher, J. D. (2019).
Deep Learning.
MIT Press.